BA2: Digital Korea

Week 9: Sentiment Analysis

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Today's Agenda

1. Week 8 review & check-in
2. What is sentiment analysis?
3. Dictionary-based approaches
4. Sentiment arithmetic: word + word + word = ?
5. Korean sentiment dictionaries
6. Application: Moon Jae-in's tweets
7. Demo: Sentiment analysis in Orange Data Mining
8. Looking ahead

Review & Check-In

From Meaning to Feeling

Week 8 asked what words *mean*; Week 9 asks how they *feel*

Last week (embeddings): words as vectors capture **meaning**.

한국 − 서울 + 도쿄 ≈ **일본**

This week (sentiment): words carry **feeling**, and the researcher decides in advance which feelings.

Two different questions

Embeddings: *what does this word mean in context?*
Sentiment: *does this word feel positive or negative, yes or no?*

What Is Sentiment Analysis?

What Is Sentiment Analysis?

Sentiment analysis = determining whether a text expresses a positive, negative, or neutral attitude.

Examples in everyday life

- Product reviews (5 stars?)
- Social media posts (happy? angry?)
- News coverage (favorable? critical?)
- Political speeches (hopeful? alarming?)

Why it matters for research

- Track public opinion over time
- Compare tone across leaders
- Detect emotional framing
- Measure media bias

In computational text analysis

Sentiment analysis is a form of **measurement**: we assign a numeric value (valence) to text that captures its emotional direction.

Two Main Approaches

Dictionary-based (this week)

- Humans create word lists
- Each word gets a sentiment score
- Computer looks up and sums scores
- Transparent and interpretable
- No training data needed

Also called: lexicon-based, rule-based

Machine learning-based

- Train a model on labeled examples
- Model learns patterns from data
- Can capture context and nuance
- Requires training data
- Less transparent ("black box")

Also called: supervised classification

This week's focus

Dictionary-based methods — where the researcher defines the rules and the computer applies them at scale.

Step 1: How a Word Gets a Value

A sentiment dictionary is just two word lists, written by humans

positive.txt		
감사	행복	희망
사랑	축하	발전
...		

negative.txt		
위기	걱정	파괴
고통	어려움	가난
...		

How the words got there: linguists and annotators read Korean text and decide, word by word, which ones carry positive feeling and which negative. They write the decisions down. That is the dictionary.

What this means in practice

A word is **positive** if it appears in `positive.txt`, **negative** if in `negative.txt`, otherwise it doesn't count. Two yes/no questions per word — no grey zone, no intensity.

Step 2: Matching Words in a Document

For every content word, ask two yes/no questions

Input: 국민 여러분께 감사드리며 희망을 함께 만들어 갑시다
(People of the nation, thank you — let's build hope together)

After tokenizing with Kiwi, we keep the content words (nouns, verbs, adjectives ≥ 2 chars): 국민 감사 드리 희망 만들

Word	In positive.txt?	In negative.txt?	Result
국민 (people)	no	no	no match
감사 (thanks)	yes	no	positive hit
드리 (to give)	no	no	no match
희망 (hope)	yes	no	positive hit
만들 (to make)	no	no	no match

Totals for this tweet: 2 positive hits, 0 negative hits, across 5 content words.

Step 3: Turning Matches Into a Score

One simple formula — used by Orange, by R, and by the interactive

$$\text{score} = 100 \times \frac{\text{positive hits} - \text{negative hits}}{\text{number of content words}}$$

Applied to our 5-word example:

$$100 \times \frac{2 - 0}{5} = +40.00$$

Why divide by length?

Long tweets would always look more extreme than short ones (more words, more chances to match). Dividing by length turns the score into a **rate** — share matched positive minus share matched negative, $\times 100$.

This is the only formula at play in Week 9 — every tweet score comes from this line.

Sentiment Arithmetic

Sentiment Arithmetic: Word + Word + Word = ?

Last week: **word + word = meaning** (embedding analogies)
This week: **word + word + word = feeling** (sentiment scoring)

희망	+	경제	+	감사	=	Positive
$+1$		0		$+1$		$+2$
위기	+	경제	+	파괴	=	Negative
-1		0		-1		-2

Each positive word adds $+1$, each negative adds -1, words not in the dictionary are ignored.
The sum is the tweet's raw sentiment score.

Your Turn: Guess the Sentiment

What is the sentiment of each combination?

1. 행복 (happy) + 사랑 (love) + 희망 (hope) = **?**
2. 공포 (fear) + 고통 (pain) + 슬픔 (sadness) = **?**
3. 감사 (thanks) + 어려움 (difficulty) + 희망 (hope) = **?**
4. 기쁨 (joy) + 위기 (crisis) + 신뢰 (trust) = **?**

Think about it

Cases 3 and 4 are **mixed**. This is where dictionary methods get interesting — and where their limitations start to show.

Sentiment Arithmetic: Answers

#	Words	Hits	Result
1	행복 + 사랑 + 희망	3 pos, 0 neg	**Positive** $(+3)$
2	공포 + 고통 + 슬픔	0 pos, 3 neg	**Negative** (-3)
3	감사 + 어려움 + 희망	2 pos, 1 neg	**Positive** $(+1)$
4	기쁨 + 위기 + 신뢰	2 pos, 1 neg	**Positive** $(+1)$

Key insight

Dictionary methods treat every word independently. They don't understand *context*: "overcome a crisis" and "cause a crisis" would score the same for 위기.

Korean Sentiment Dictionaries

Korean Sentiment Dictionaries: The Landscape

Several research dictionaries are publicly available for Korean

Dictionary	Size	Scale	Notes
KNU (Park et al. 2018)	14,854	-2 to $+2$	**Course dictionary (what we use)**
KOSAC (Jang et al. 2013)	~11,275	5-class + intensity	SNU manual annotation on news
Chen & Skiena (2014)	44 langs	binary	Orange's built-in Multilingual option
NRC Emotion Lexicon	~14,000	8 emotions $+\pm$	Cross-lingual, MT Korean
KSenticNet	5,465	continuous	Neural / concept-level

Why KNU?

Published Korean sentiment research standard. Simple word lists (positive, negative) that we load into Orange via Custom Dictionary. You can open the files in any text editor and read the words.

One Method, Two Places to See It

Same pipeline in Orange and in the interactive — numbers match

In Orange (you build this)

- File → Corpus
- Python Script (Kiwi)
- Corpus (re-map)
- Sentiment Analysis (Custom Dict)
- Box Plot

Five widgets, one script, loads KNU positive/negative files.

In the interactive (6 panels)

1. The Corpus — tweets per year
2. Dictionary Scoring — one tweet at a time
3. Score Distribution — histogram
4. By Period — box plot
5. Over Time — timeline + events
6. Explore Tweets — browse extremes

Same Kiwi + KNU pipeline, pre-computed.

Why both?

Orange gives you the tool. The interactive opens it up so you can see what the tool is doing to your tweets.

The KNU Korean Sentiment Lexicon

Published by Park et al. (2018) at Kunsan National University

KNU annotators read Korean dictionary entries, emoticons, neologisms, and other lexicons, and decided word by word which carry positive feeling and which carry negative. We use the two plain word lists they published:

File	Words	Contents
`positive.txt`	4,868	Words KNU labelled positive
`negative.txt`	9,824	Words KNU labelled negative

Note on the files

These are KNU's own `pos_pol_word.txt` / `neg_pol_word.txt` with the header stripped — no intensity scores, no modifications by us. KNU also publishes a richer `SentiWord_Dict.txt` with -2 to $+2$ intensity; we ignore it because Orange's Custom Dictionary widget only reads plain word lists.

What's in the KNU Word Lists?

A peek inside — these are the words the widget will look for

`positive.txt` (samples)

- 감사 (gratitude)
- 행복 (happiness)
- 축하 (celebration)
- 사랑 (love)
- 희망 (hope)
- 발전 (development)

`negative.txt` (samples)

- 가난 (poverty)
- 파괴 (destruction)
- 위기 (crisis)
- 걱정 (worry)
- 어려움 (hardship)
- 고통 (suffering)

You can open the files yourself

Plain text, one word per line, no header. There is no magic behind the dictionary — it is just a list.

Orange Under the Hood

Orange's Sentiment Analysis Widget: Two Options

Same scoring formula, different word lists

Option 1: Multilingual (built-in)

- Method = Multilingual, Language = Korean
- Orange ships with its own Korean word lists (Chen & Skiena 2014)
- Zero setup, zero files to download

Quickest way to get a number out.

Option 2: Custom Dictionary

- Method = Custom Dictionary
- *You* provide two files: positive word list and negative word list
- We provide KNU's `positive.txt` and `negative.txt`

This is what we'll use.

Same formula, different words

Both options apply the **same scoring formula**: tokenize, count hits in each list, divide by length. The only thing that changes is *which word lists* the widget looks in.

Under the Hood: What the Widget Actually Does

No black box — the same 3 steps as before, now inside Orange

When you connect a Corpus to Sentiment Analysis (Custom Dictionary + `positive.txt` / `negative.txt`), here's what happens to **every document**:

1. Tokenize the document — Kiwi already did this in our pipeline.

↓

2. Look each token up — is it in `positive.txt`? In `negative.txt`? Count unique hits per document.

↓

3. Apply the formula — score $= 100 \times (\text{pos} - \text{neg})/\text{num tokens}$.

What the widget outputs

A new `sentiment` column on the corpus, one number per tweet. Connect that output to a Box Plot, Distributions, or Line Plot. The widget adds a number; downstream widgets make pictures.

Application: Moon Jae-in's Tweets

Case Study: Moon Jae-in on Twitter

3,148 tweets from @moonriver365 (2012–2020), scored with KNU

The corpus: @moonriver365

- 3,148 tweets (2012–2020), 9 columns
- Metadata: date, favorites, retweets
- Most active during the 2012 campaign (1,427 tweets)
- Tweeting declined sharply after he took office in 2017

Moon Jae-in (문재인) — 19th President (2017–2022)

- Key topics: inter-Korean relations, COVID-19, Japan trade dispute, democracy

Research questions:

1. Is the overall tone positive or negative?
2. Does sentiment change across the three periods?
3. Do different topics carry different sentiment?
4. How does sentiment relate to real-world events?

Three Political Periods: My Editorial Split

The `period3` column is my grouping, not an official classification

Period	Dates	Tweets	What's happening
Pre-presidency	2012-01 to 2016-11	1,973	Opposition leader — 2012 campaign, DP chairmanship, legislative politics
Transition	2016-12 to 2017-05	393	Park Geun-hye impeachment crisis through Moon's campaign and inauguration
Presidency	2017-05 to 2020-06	782	In office — inter-Korean summits (2018), Japan trade dispute (2019), COVID-19 (2020)

A research choice, not a fact

I drew the boundaries at the impeachment vote and the inauguration. A different researcher might cut the timeline differently — by election cycle, Twitter-volume phase, topic. **The categories are part of your analysis, not a given.**

Sentiment Arithmetic with Real Tweets

A purely positive tweet: celebrating a birthday

Tweet (2018-01-24, presidency):
생일 축하, 고맙습니다. 생일을 챙기지 않는 삶을 살아왔는데 ··· 진심으로 감사드립니다.
(Thank you for the birthday wishes. I've lived without celebrating birthdays ... I am truly grateful.)

Matched word	In dictionary
축하 (celebration)	positive
감사 (gratitude)	positive

2 positive matches, 0 negative. Orange's formula: $100 \times (2 - 0)/14$ tokens $= +14.3$.

Sentiment Arithmetic with Real Tweets

A mixed-sentiment tweet: grief, solidarity, hope

Tweet (2020-03-30, presidency):
어려움 속에서도 서로를 격려해가며 신뢰와 협력으로 재난을 이겨가고 있는 국민들께 한없는 존경과 감사를 드립니다.
(Even amid hardship, I offer deepest respect and gratitude to the people enduring this disaster with trust and cooperation.)

Matched word	In dictionary
신뢰 (trust)	positive
존경 (respect)	positive
감사 (gratitude)	positive
어려움 (hardship)	negative
재난 (disaster)	negative

3 positive, 2 negative in 20 tokens: $100 \times (3-2)/20 = +5.0$. The tweet leans positive because gratitude and respect outweigh the hardship words.

Moment in Time: The Impeachment Vote

Tweet (2016-12-09, transition, 4,445 favorites):

국민이 이겼습니다. 대통령 탄핵은 끝이 아니라 시작입니다.

(The people have won. The presidential impeachment is not the end but the beginning.)

Dictionary scoring:

- Matched words: **0**
- 국민, 이기, 대통령, 탄핵, 시작 — none of these are in KNU
- Score: **0** (neutral)

Context:

- Posted day after the impeachment vote
- Tone is triumphant
- Dictionary says "neutral" — the method misses the political significance entirely

Critical Reading: What the Dictionary Misses

Tweet (2019-08-02, presidency, 19,110 favorites):

우리는 다시는 일본에게 지지 않을 것입니다. 수많은 역경을 이겨내고 ⋯ 어려움을 극복할 역량이 있습니다.

(We will never again lose to Japan. We have overcome many adversities ... we have the capability to overcome difficulties.)

Dictionary scoring:

- 어려움 (difficulty) $\rightarrow$ **negative**
- 역경 (adversity) $\rightarrow$ **negative**
- 극복, 역량, 일본 — not in KNU
- Score: -16.67 (two negative hits in 12 content tokens)

Human reading:

- Tone is defiant and assertive
- Negative *toward Japan* specifically
- Positive *self-framing* (national strength)
- The dictionary sees just “difficulty”

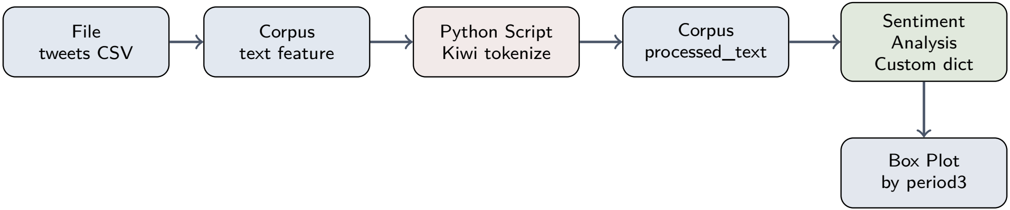
Lesson

Dictionary-based sentiment measures **word-level valence**. It doesn’t capture the **target** (negative toward whom?) or the **rhetorical purpose** (rallying, not despairing).

Minimum Orange Workflow

Minimum Workflow for the Assignment

Five widgets plus one script — the minimum for class



- **File / Corpus**: load `moon_twitter.csv`, text feature = `text`.
- **Python Script**: paste `sentiment_preprocessing_*.py`. Writes a `processed_text` column.
- **Second Corpus**: re-map used text feature to `processed_text`.
- **Sentiment Analysis**: Method = Custom Dictionary; load `positive.txt` and `negative.txt`.
- **Box Plot**: variable = sentiment, subgroup = `period3`.

In class I'll demo more views off the same Sentiment Analysis output.

Interpreting Your Results

Once you have sentiment scores, ask:

1. **Distribution:** What is the overall shape? Mostly positive? Skewed?
2. **Outliers:** Which documents scored extremely high or low? Do they make sense when you read them?
3. **Patterns:** Does sentiment correlate with time, topic, or another variable?
4. **Validation:** Read 10–20 documents across the score range. Does the algorithm's judgment match yours?

Always validate!

Grimmer, Roberts & Stewart: "All quantitative models of language are wrong — but some are useful" (Ch. 2). Validation is not optional.

Looking Ahead

For Next Week

Week 10: Topic Modeling (LDA)

- Latent Dirichlet Allocation (LDA)
- Discovering hidden thematic structure in a corpus
- Choosing the number of topics
- Interpreting and labeling topics

Recommended reading:

- Grimmer, Roberts & Stewart — Chapter 13: Topic Models

For Next Week

Interactive exercise, assignment, and preparation

Interactive exercise:

- Explore the **Sentiment Analysis: Moon Jae-in's Tweets** interactive on the course website
- Step through all 6 panels to see exactly what Orange is doing with your tweets

Assignment:

- Build the 5-widget Orange workflow on `moon_twitter.csv`: **File** → **Corpus** → **Python Script** (Kiwi) → **Corpus** → **Sentiment Analysis** (Custom Dict with `positive.txt`, `negative.txt`) → **Box Plot**
- Write 2–3 sentences: which period is most positive, and which matched words from the interactive explain why?

Preparation:

- Download `moon_twitter.csv`, `sentiment_preprocessing_*.py`, `positive.txt`, `negative.txt` from the **Data & Scripts** page